Uncertainty & Bayesian Networks

(1)

[a] Valid probability distribution

- $0 \leq P(\omega) \leq 1$ ✓ $\quad 0 \leq 0.56, 0.18, 0.14, 0.12 \leq 1$
- $\sum_{\omega} P(\omega) = 1$ ✓ $\quad 0.56 + 0.14 + 0.18 + 0.12 = 1$

[b] $P(dusty) = 0.12 + 0.18 = 0.3000$

$P(late) = 0.14 + 0.18 = 0.3200$

[c] $P(late \mid dusty)$

$$\frac{P(late \wedge dusty)}{P(dusty)} = \frac{0.18}{0.30} = 0.6000$$

$P(late \mid clear)$

$$\frac{P(late \wedge clear)}{P(clear)} = \frac{0.14}{0.56+0.14} = 0.2000$$

[d] Independent or not? $P(A \cap B) = P(A)P(B)$

$P(late \wedge dusty) = P(late) \times P(dusty)$

$0.18 = 0.32 \times 0.30$

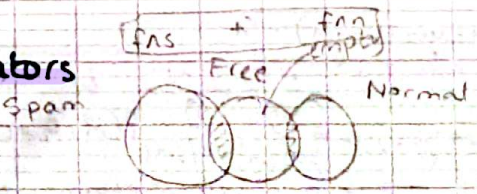
$0.18 \neq 0.9600$

Weather & bus are not independent.

[d] $P(dusty \mid late) \overset{??}{=} P(late \mid dusty)$

$$\frac{0.18}{0.14+0.18} \Rightarrow 0.5625 \neq 0.6000$$

Not equal because they're divided by different denominators



(2)

[a] $P(s) = 0.3 \quad P(n) = 0.7$

$P(f \mid s) = 0.6 \rightarrow 0.6 = \frac{f \wedge s}{0.3} \rightarrow P(f \wedge s) = 0.18$

$P(f \mid n) = 0.05 \rightarrow 0.05 = \frac{f \wedge n}{0.7} \rightarrow P(f \wedge n) = 0.035$

$\} P(f) = 0.18 + 0.035 = 0.2150$

[b] $P(s \mid f) = ?$

$$P(s \mid f) = \frac{P(s \wedge f)}{P(f)} = \frac{0.18}{0.2150} = 0.8372$$

[c] $P(s \mid f, \omega) = ?$

$P(\omega \mid s) = 0.4 \quad P(\omega \mid n) = 0.02$

$$= \frac{[P(s) \cdot P(f \mid s) \cdot P(\omega \mid s)]}{P(f \mid \omega) - \text{normal or spam}} = 0.3 \times 0.6 \times 0.4 = 0.0720$$

$\rightarrow P(f, \omega \mid n) = P(n) \cdot P(f \mid n) \cdot P(\omega \mid n) = 0.7 \times 0.05 \times 0.02 = 0.0007$

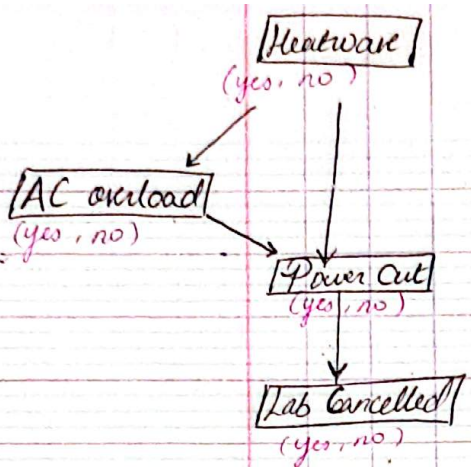
$\rightarrow P(f \mid \omega) = P(f, \omega \mid n) + P(f, \omega \mid s) = 0.0007 + 0.0720 = 0.0727$

$$P(s \mid f, \omega) = \frac{0.0720}{0.0727} = 0.9904$$

[d] $P(s)$ goes ↑ or ↓? (0.40)

$P(s)$ increases from 83% to 99% when adding the word winner because it appears more frequently than in normal emails (0.02)

③ (a)



(b) $P(H,A,C,L) = P(H) \cdot P(A|H) \cdot P(C|A,H) \cdot P(L|C)$

$= 0.3 \times 0.6 \times 0.7 \times 0.8 = 0.1008$

(c) $P(H) \rightarrow$ yes $= 0.3 \rightarrow 1$

$P(A|H) \rightarrow Y,Y$ or $Y,N \rightarrow 2$

$P(C|H,A) \rightarrow H=2, A=2 \rightarrow 2\times2=4$

$P(L|C) \rightarrow C=2$

Y,Y,Y Y,N,Y Y,Y,N Y,N,N

L|C Y,Y or Y,N

The table needs 9 rows.

(d) $P(H=yes, A=no, C=yes, L=yes)$

$P(H) \times P(A|H) \times P(C|A,H) \times P(C|L)$

$0.3 \times 1-0.6 \times 0.2 \times 0.8 = 0.0192$

(e) $P(H=yes|C=yes)$

$$P(H|C) = \frac{P(C|H)\,P(H)}{P(C)}$$

$P(C=yes|H=yes) = [P(C=yes|H=y, A=y) \cdot P(A=y|H=y)] + [P(C=yes|H=yes, A=no) \cdot P(A=no|H=yes)] = 0.5000$ (0.7, 0.6, 0.2, 0.4)

$P(C=yes|H=no) = [P(C=yes|H=no, A=y) \cdot P(A=y|H=no)] + P(C=yes|H=no, A=no) \cdot P(A=no|H=no)] = 0.0950$ (0.5, 0.1, 0.05, 0.9)

$P(C|H) = 0.5$ $P(H) = 0.3$ $P(C) = (0.5 \times 3) + (0.095 \times 2) = 0.2165$

$\hookrightarrow P(H=yes|C=yes) = (0.5 \times 0.3)/0.2165 = 0.6928$

$P(H=yes|C=yes, L=yes) = 0.6928$ because L depends directly on C only & hence provid no additional information.

④

(a) $P(H=yes|L=yes) = ?$

Query (X) → H Evidence (E) → L Hidden (Y) → A, C

$$P(H|L=yes) = \alpha \sum_{a\in\{y,n\}} \sum_{c\in\{y,n\}} P(H, A=a, C=c, L=yes)$$

H = yes → 0.1008 + 0.0054 + 0.0192 + 0.0096 = 0.1350

a=y, c=y : $0.3 \times 0.6 \times 0.7 \times 0.8 = 0.1008$

a=y, c=n : $0.3 \times 0.6 \times 0.3 \times 0.1 = 0.0054$

a=n, c=y : $0.3 \times 0.4 \times 0.2 \times 0.8 = 0.0192$

a=n, c=n : $0.3 \times 0.4 \times 0.8 \times 0.1 = 0.0096$

H = no → 0.028 + 0.0035 + 0.0252 + 0.05985 = 0.1

a=y, c=y $0.7 \times 0.1 \times 0.5 \times 0.8 = 0.0280$

a=y, c=n $0.7 \times 0.1 \times 0.5 \times 0.1 = 0.0035$

a=n, c=y $0.7 \times 0.9 \times 0.05 \times 0.8 = 0.0252$

a=n, c=n $0.7 \times 0.9 \times 0.95 \times 0.1 = 0.05985$

Sum = 0.1350 + 0.11655 = 0.25155

$P(H=yes|L=yes) = 0.1350/0.025155 = 0.5367$

(b) $P(A=yes, L=yes) = ?$

Query = A, Evidence = L, Hidden = H, C

A = yes → 0.1062 + 0.0315 = 0.1377

H=y, a=y → 0.1008 + 0.0054 = 0.1062

H=n, a=y → 0.028 + 0.0035 = 0.0315

A=no → 0.0288 + 0.08505 = 0.11385

H=y, a=y → 0.0192 + 0.0096 = 0.0288

H=n, a=n → 0.0252 + 0.05985 = 0.08505

Sum = 0.1377 + 0.11385 = 0.25155

$P(A=yes|L=yes) = \frac{0.1377}{0.25155} = 0.5474$

c. $P(H=yes \mid L=yes, A=no)$

$H=yes\ (A=no, L=yes) = 0.0192 + 0.0096 = 0.0288$

$H=no\ (A=no, L=yes) = 0.0252 + 0.05985 = 0.08505$

$P(H=y \mid L=y, A=n) = \frac{0.0288}{0.11385} = 0.2530$

The probability of a heatwave decreased. This is because H & A are the 2 factors influencing C, which influences L. If A is ruled out, as in the last equation, it reduces the likelihood of a heatwave being responsible

⑤

a. 1 $P(L=yes)$ keep $\{2,4,6,7,12\}$ throw $\{1,3,5,8,9,10,11\}$

$P(H=yes \mid L=yes) = \{2,7\} = \frac{2}{5} = 0.4000$

2 Fraction of samples thrown away $= 7/12 = 58.33\%$

Rejection sampling is inefficient when the discarded samples are high, because it generates many samples just for them to get discarded

b. 1 $w_i = P(L=yes \mid C_i)$ $P(L=yes \mid C=yes) = 0.8$ $P(L=yes \mid C=no) = 0.1$

Sample 1 $\Rightarrow$ C=no $w_1=0.1$ Sample 2 $\Rightarrow$ c=yes $w_2=0.8$ Sample 3 $\Rightarrow$ c=no $w_3=0.1$

Sample 4 $\Rightarrow$ C=no $w_4=0.1$ Sample 5 $\Rightarrow$ c=yes $w_5=0.8$ Sample 6 $\Rightarrow$ c=yes $w_6=0.8$

1 Estimate $P(H=yes \mid L=yes)$

$= \frac{0.8+0.8+0.1}{1.7+0.1+0.8+0.1} \begin{matrix}[H=yes] \\ [sum]\end{matrix} = \frac{1.7}{2.7} = 0.6296$

c. The answers from a (0.4000) & b (0.6296) are different from the exact answer due to their small sample sizes ($S=12$, $S=6$). The answers can be made closer by giving them the same sample size or increasing the sample size substantially to where they converge.